

BEHAVIOR MODIFICATION & COMPLIANCE

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Lecture Learning Outcomes

- Outline theories of behavior change
- Describe health belief model and clinician's approach to behavior change

Reference:

Chapter 2: Ramseier C, Suvan JE, editors. Health behavior change in the dental practice. John Wiley & Sons; 2011 Jun 9.

Introduction

- ✓ **Behavior modification** refers to structured strategies aimed at changing unhealthy habits into healthier ones.
- ✓ In oral health promotion, it is crucial for ensuring lasting compliance and improving patient outcomes across populations.

Example of a patient behavior

Mrs. K is treated by a dentist. Her teeth are cleaned by the dentist, and she is taught how to brush.

For the first few weeks she brushes as explained.

Slowly she starts to brush wrongly.

One day she finds that her gums are bleeding.

She is now afraid and does NOT want to go to the dentist and cancels her appointment.

Compliance

- Patient's behaviors (in terms of taking medication, following diets, or executing life style changes) match with healthcare providers' recommendations for health and medical advice

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Non-compliance:

- Failure or refusal to comply.
- In medicine, the term non-compliance is commonly used in regard to a patient **who does not take a prescribed medication or follow a prescribed course of treatment.**
- A person who demonstrates non-compliance is said to be noncompliant.



Non-compliance: Causes

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▶ **Language barrier**

- ▶ Patients might not understand how to take medication, perform oral hygiene techniques, or follow pre- and post-operative care guidelines

▶ **Low education level**

- ▶ Limited understanding of oral health
- ▶ Prioritize immediate pain relief over long-term preventive measures

▶ **Poor dentist-patient interaction**

- ▶ less likely to follow recommendations.

▶ **System related obstacles**

- ▶ Financial constraints
- ▶ Limited access to care
- ▶ Complex administrative processes e.g. insurance paperwork
- ▶ Inconvenient appointment times



**Why
behavior
change is
important?**

- **Improves Oral Health Outcomes**
- **Prevents Disease Progression**
- **Enhances Treatment Effectiveness**
- **Reduces Need for Further Intervention**
- **Improves Overall Health**
- **Increases Patient Satisfaction & Confidence**
- **Strengthenes Dentist-Patient Relationship**

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- **Requires adopting new habits and new lifestyle patterns**



What is Ambivalence?

refers to the state of having **mixed feelings or contradictory ideas** about adopting or maintaining a particular **oral health behavior**.



The **desire** for good oral health **is present**, but the perceived **barriers and inconvenience create resistance**. Some days they will follow desired behavior and other days they don't

Theories and Mechanisms of Motivation, Compliance & Behavior

1. Foundations of Health Behavior Change
2. Theory of planned behavior
3. Social cognitive theory
4. Patient compliance mechanisms
5. The health belief model

1. Foundations of Health Behavior Change

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- **Motivation** of dental behavior can be
 - **Intrinsic** (driven by personal values)
 - A child brushes twice daily because he/she values having a clean mouth and admires their dentist.
 - A dental student does flossing regularly after learning about periodontal disease in class and wanting to model good oral health.
 - **Extrinsic** (influenced by external rewards)
 - A parents rewards their child with a sticker for brushing every night..
- **Habit Formation Through Reinforcement & Repetition**
 - **Positive reinforcement** during recall visits (e.g., “Your gums look great!”) encourages continued flossing

2. Theory of Planned Behavior:

- ▶ A patient's **likelihood of following dental advice** is shaped by their
 - ▶ **Attitudes** (e.g., belief in the value of flossing),
 - ▶ **Social norms** (e.g., family expectations about oral hygiene),
 - ▶ **Perceived behavioral control** (e.g., confidence in their ability to maintain a brushing routine).

3. Social Cognitive Theory:

- **Self-efficacy and role modeling** influence how people **adopt and maintain oral health habits.**
- **children imitate parents or dental professionals** who demonstrate proper oral hygiene habits.

4. Patient Compliance Mechanisms:

- Compliance can be achieved by 3 factors
 - **Cognitive** (understanding the importance of oral care),
 - **Emotional** (fear of pain or motivation to avoid cavities),
 - **Social** (influence of family, culture, or peer norms).
- Recognizing barriers such as low literacy, cultural beliefs, and dental anxiety is essential for realistic interventions.

5. THE HEALTH BELIEF MODEL.

Patients are more likely to **adopt preventive oral health behaviors** when they:

- 1. Perceive a personal threat** to their dental health—such as recognizing the risk of developing caries or periodontal disease.
- 2. Believe in the efficacy of the recommended intervention**—for example, trusting that fluoride application or improved brushing will effectively reduce that risk.
- 3. Feel capable of overcoming barriers**—whether those are financial, psychological (e.g., dental anxiety), or practical (e.g., lack of time or access).

This model is useful** when **designing patient education strategies, tailoring motivational interviewing, or framing public health campaigns around caries prevention and oral hygiene.

Factors that influence Behavior & Motivation

► Cues to action:

- A school dental screening or a parent's reminder **prompts a child to brush regularly.**

► Demographics (gender):

- Studies show that **females** may be more likely to **seek preventive dental care than males.**

► Psychological characteristics

► Personality:

- **Conscientious individuals** often **adhere better to oral hygiene routines.**

► Peer group pressure:

- **Teens** may **adopt whitening or aesthetic treatments** based on peer influence.

How does the behavior change?

- Change can happen **naturally** in everyday life
- **Intrinsic motivation** affects patient behavior
- People change when they are **ready**
- **Ambivalence** is part of the process
- Patient behavior change **happens outside the treatment room**

Change can happen naturally in everyday life

- ▶ A 45-year-old male with a history of **chronic smoking** presented with clinically evident **gingival deterioration**. Despite repeated **professional advice** to stop smoking, he **remained non-compliant** and he **continued to smoke**.

One day, his **elder brother**, who is also a **smoker**, gets admitted to hospital with **serious lung disease**. Seeing his brother's condition, he immediately **stops smoking** and over a period his **gingival health also improve**

Intrinsic motivation affects patient behavior



Example: An **intrinsically motivated individual flosses daily because he enjoys the feeling of a clean mouth** and appreciate the long-term benefits of preventing gum disease. He doesn't need reminders or external encouragement; the satisfaction of maintaining his oral health is his own reward. This leads to consistent flossing habits.

Where and when does behavior change occur?

- ▶ Behavior change occurs **when people are ready!!**
- ▶ Behavior change occurs outside dental clinic
(**not immediate** during treatment but **over a period** and **outside**)



What can the clinician (dentist) do ?

- **Understand limitations of giving advice**
- **Interpersonal skills**
- **Agreeing on priorities**
- **Recognizing ambivalence is a part of the process**

Understand limitations of giving advice

- There are **limits** what **advice alone can benefit** because of
 - Barriers like **anxiety**
 - **Environmental and Social Factors**
 - **Underlying Beliefs and Values**
- **Conversational environment** in which the advice is given makes a significant positive impact
 - A warm, **empathetic**, and **respectful environment**
 - **Clear, simple language** that patient understands
 - **Positive and encouraging**



Agreeing on priorities

- **Involving** patient in decision making
- **Agreeing** with patient
 - what is important.



Acknowledge ambivalence



➤ Clinician must

- **accept** that there will be days when patients will be confused (**ambivalent**). This helps
 - Building Trust
 - Respecting autonomy
 - Understanding barriers
- **Actively listen to patients**
- **Focusing on the Patient's Own Reasons for Change**

Conclusion

- ▶ **Effective behavior modification leads to a shift from simply providing advice to actively engaging with patients, understanding their ambivalence, tailoring strategies to**
 - ▶ **improve treatment outcomes,**
 - ▶ **reduce disease progression,**
 - ▶ **improve overall well-being for the individuals**